Definition 5.8.1. If G is a group, $H \subset G$, and V is a representation of H, then the **induced representation** $\mathrm{Ind}_H^G V$, is the representation of G with

$$\mathrm{Ind}_H^G V = \{f : G \to V | f(hx) = \rho_V(h) f(x) \ \forall x \in G, h \in H\}$$

and the action $g(f)(x) = f(xg) \ \forall g \in G$.

Remark 5.8.2. In fact, $\mathrm{Ind}_H^G V$ is naturally isomorphic to the representation $\mathrm{Hom}_H(k[G], V)$.

Let us check that $\mathrm{Ind}_H^G V$ is well defined as a representation. Indeed, we have

$$g(f)(hx) = f(hxg) = \rho_V(h) f(xg) = \rho_V(h) g(f)(x),$$

and

$$g(g'(f))(x) = g'(f)(xg) = f(xgg') = (gg')(f)(x)$$

for any $g, g', x \in G$ and $h \in H$.

Remark 5.8.3. Notice that if we choose a representative x_σ from every right H-coset σ of G, then any $f \in \mathrm{Ind}_H^G V$ is uniquely determined by $\{f(x_\sigma)\}$.

Because of this,

$$\dim(\mathrm{Ind}_H^G V) = \dim V \cdot (G : H),$$

where $(G : H)$ is the index of H in G.

Problem 5.8.4. Check that if $K \subset H \subset G$ are groups and if V is a representation of K, then $\mathrm{Ind}_H^G \mathrm{Ind}_K^H V$ is isomorphic to $\mathrm{Ind}_K^G V$.

Exercise 5.8.5. Let $K \subset G$ be finite groups, and let $\chi : K \to \mathbb{C}^*$ be a homomorphism. Let $\mathbb{C}_\chi$ be the corresponding 1-dimensional representation of K. Let

$$e_\chi = \frac{1}{|K|} \sum_{g \in K} \chi(g)^{-1} g \in \mathbb{C}[K]$$

be the idempotent corresponding to χ. Show that the G-representation $\mathrm{Ind}_K^G \mathbb{C}_\chi$ is naturally isomorphic to $\mathbb{C}[G] e_\chi$ (with G acting by left multiplication).